

PrivML: Test Outputs

Timestamp: 2025-09-19 19:49:46
Model ID: Model 62

Dataset Path: data/Humans
Number of Images in Dataset: 7000
Training Size: 75%
Validation Size: 15%
Testing Size: 5%
Encrypted Testing Subset Size: 5%
Random Seed: 999

Plain Model Training

Epochs: 100
Batch Size: 16
Learning Rate: 0.001
Optimizer: Adam
Loss Function: MSELoss
Sample Output Count: 6

Plain Model Architecture

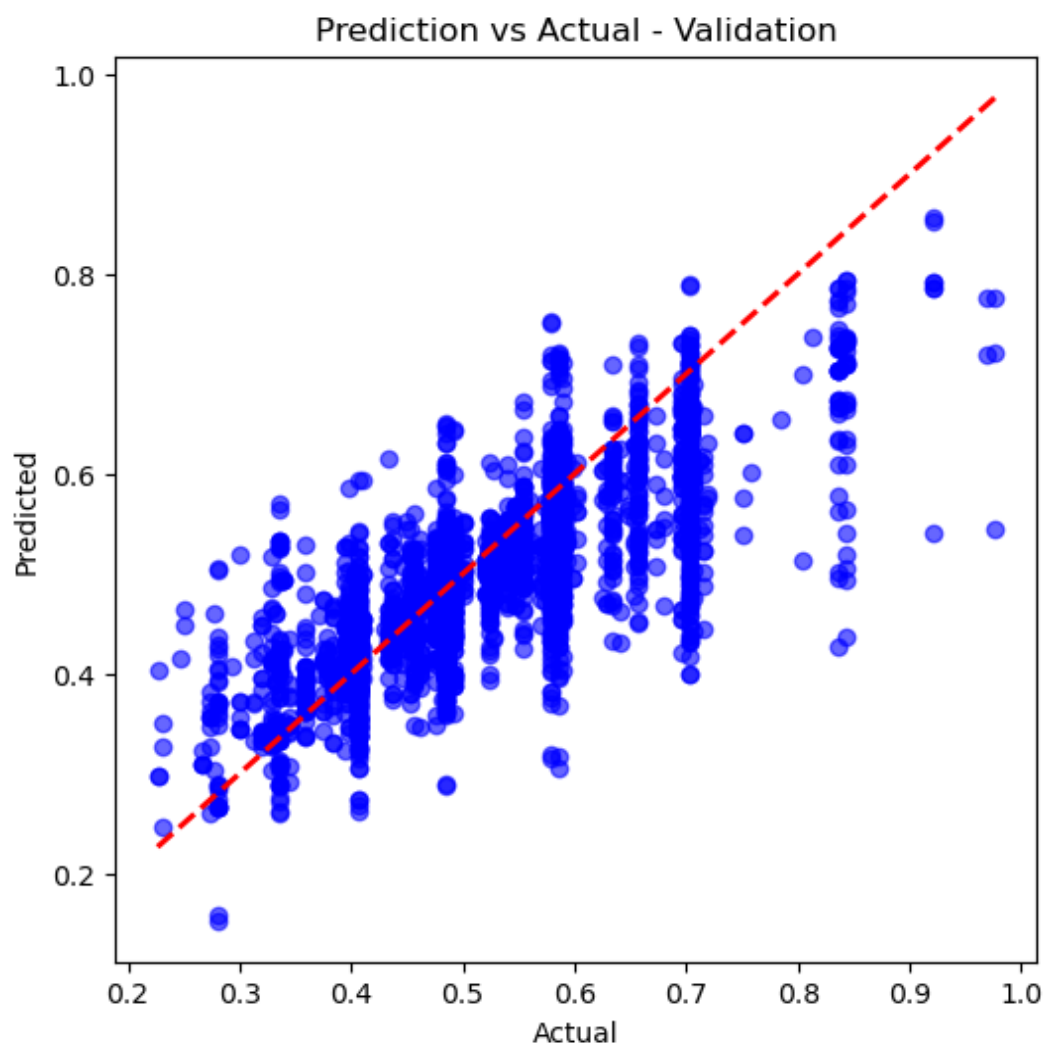
Hidden Layer Size: 512
Output Classes: 4
Conv Kernel Size: 1
Conv Stride: 2
Conv Padding: 0
Dropout Rate: 0.3

FHE Configuration

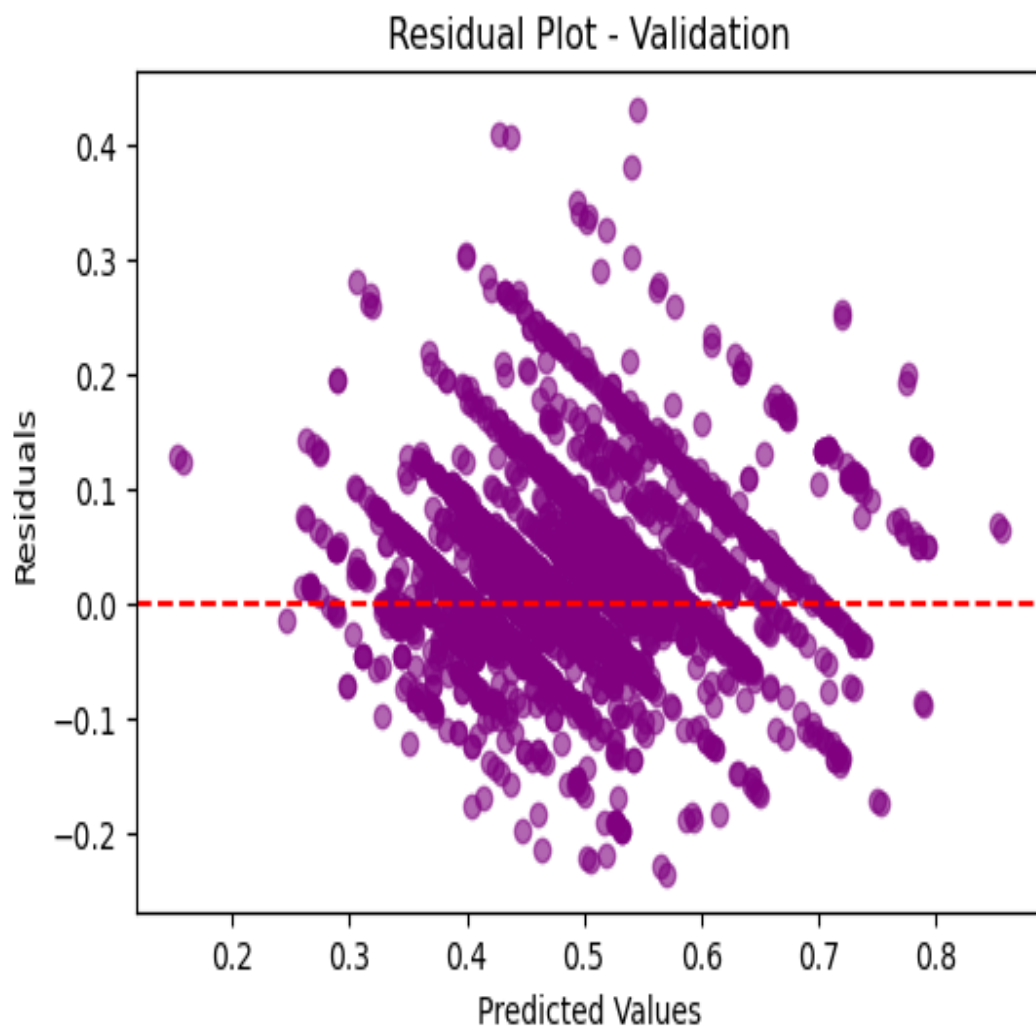
Encryption Scheme: CKKS
Scale (bits): 26
Polynomial Modulus Degree: 8192
Key Generation Level: 31
Decryption Stability Level: 31
Encrypted Loss Function: MSELoss
Coefficient Mod Bit Sizes: [31, 26, 26, 26, 26, 26, 26, 31]

Privacy-Preserving ML: ML Metrics Report

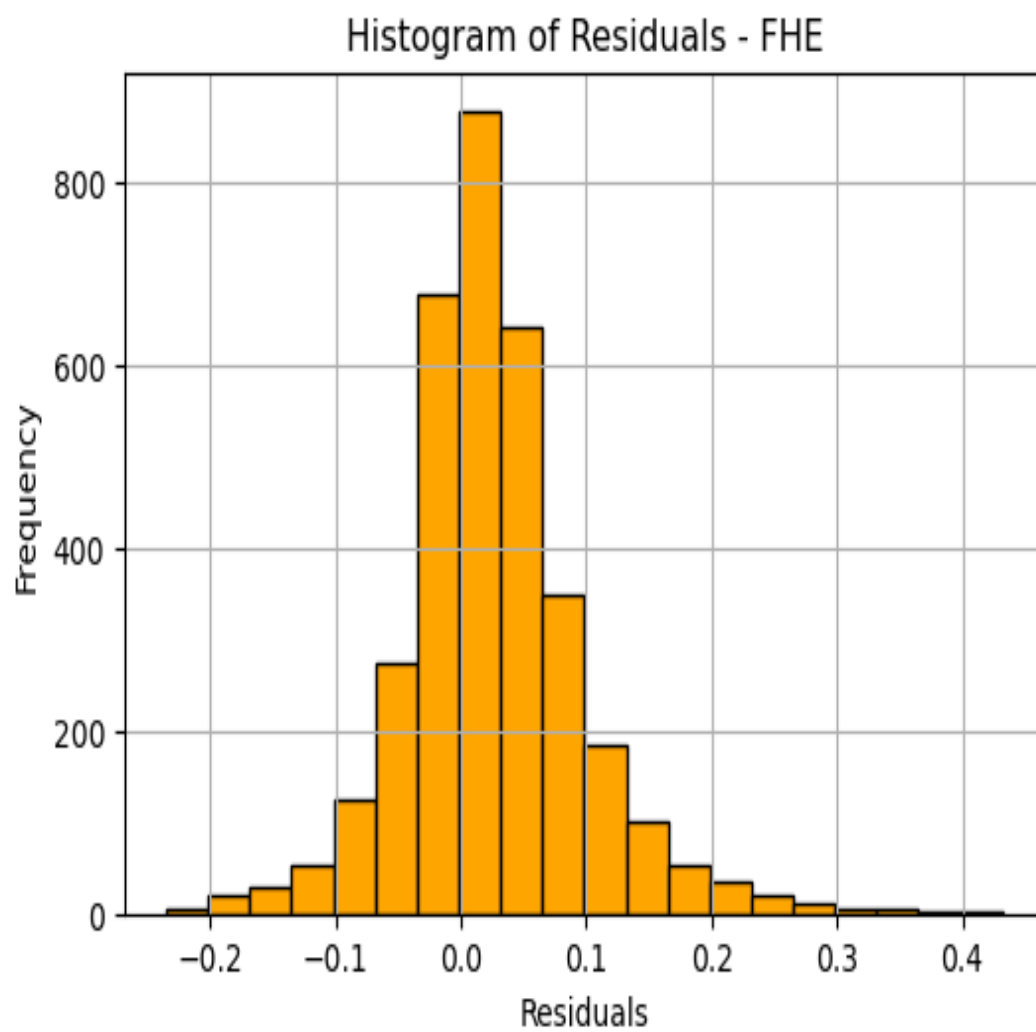
Prediction vs Plot - Validation



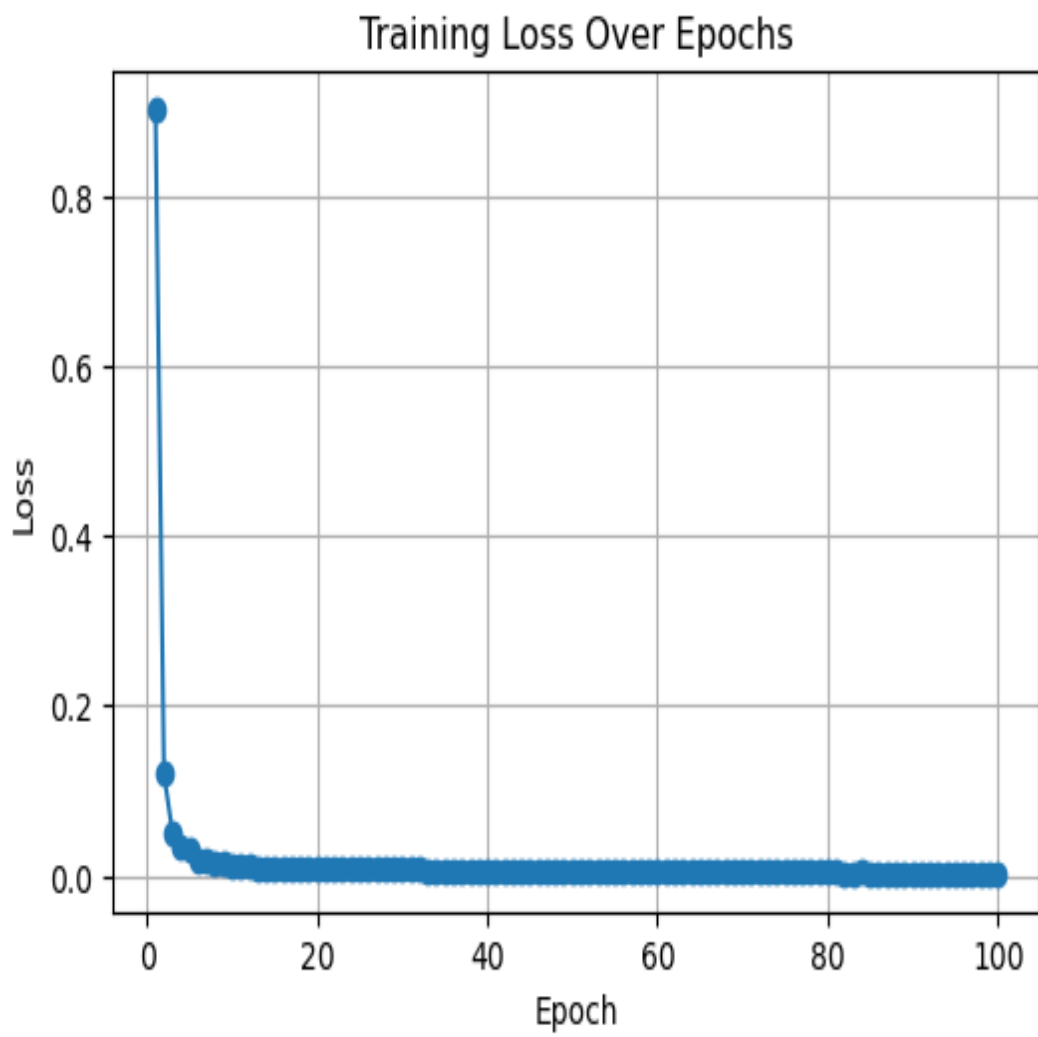
Residual Plot - Validation



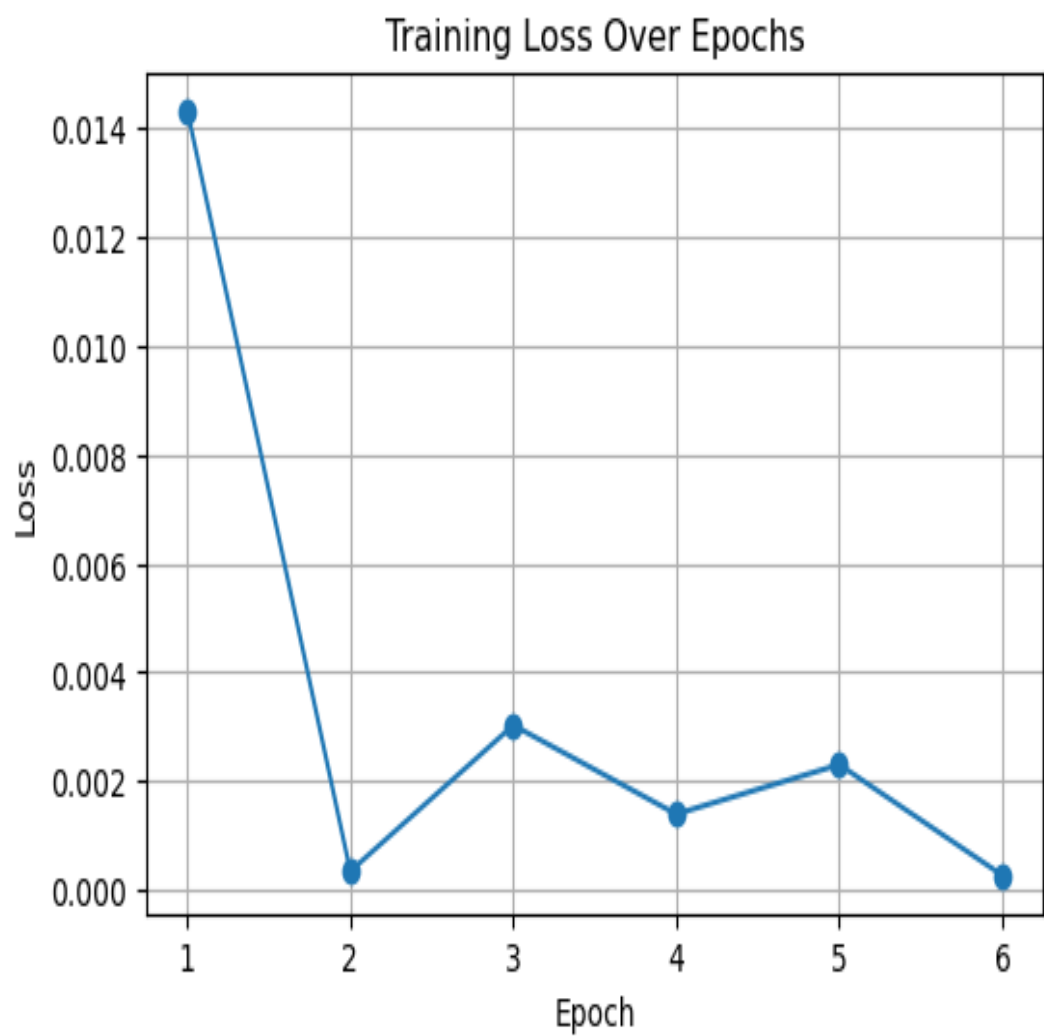
Residual Histogram - Validation



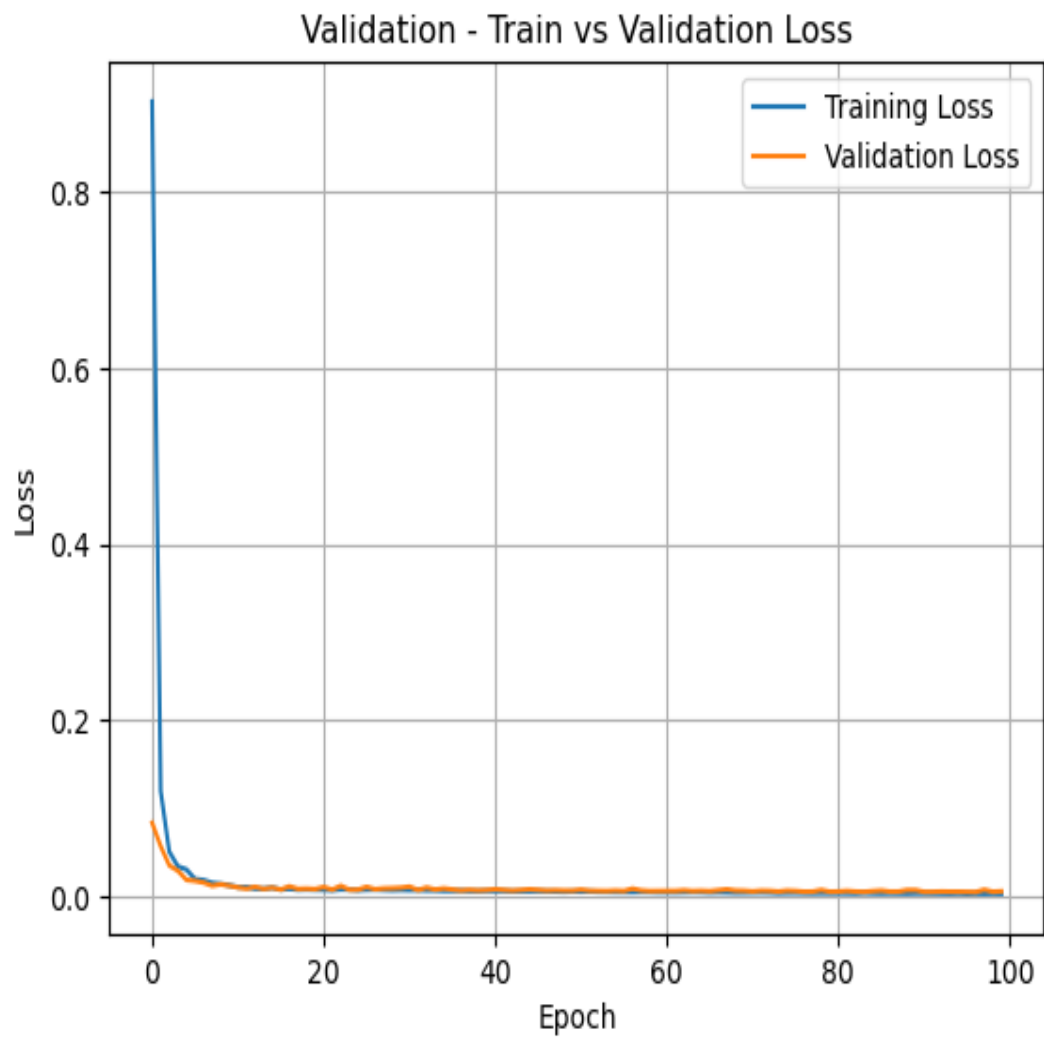
Training Loss - Validation



Training Loss - encrypted-fhe

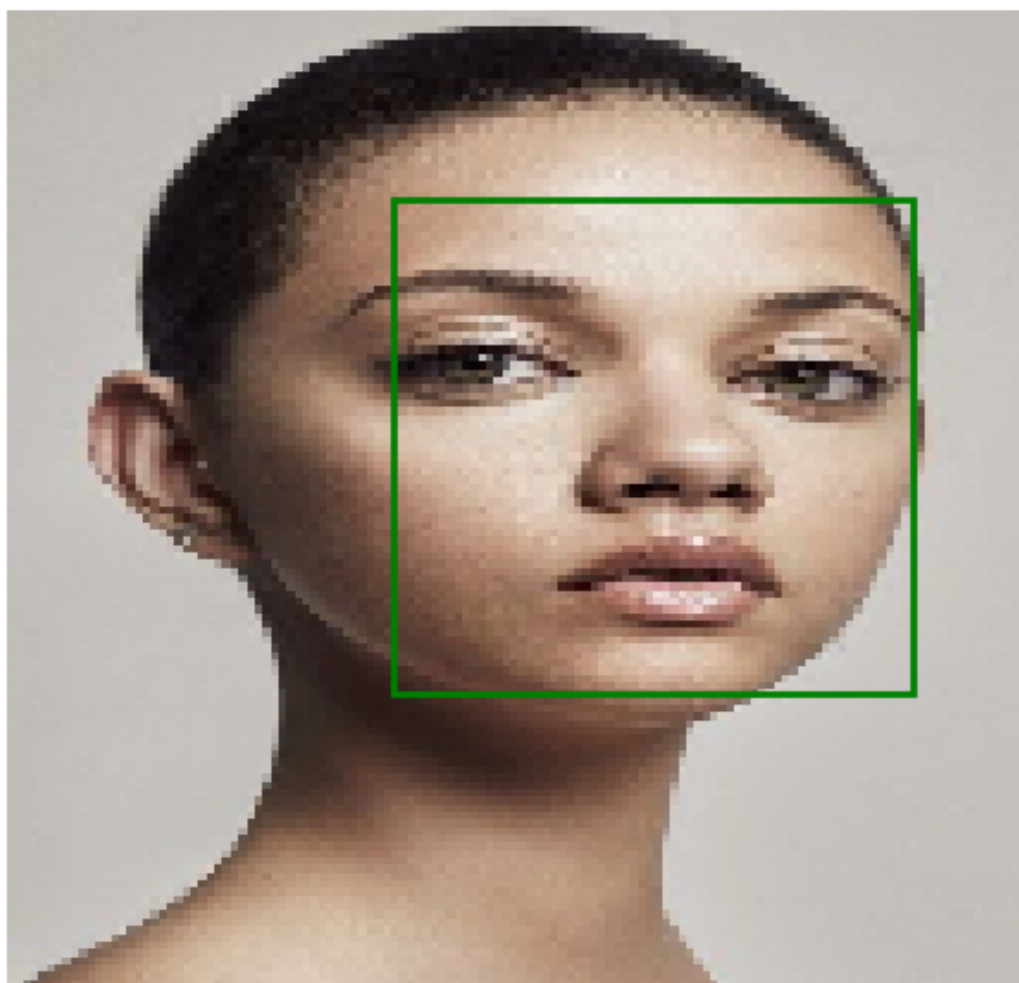


Train vs Validation Loss - Validation



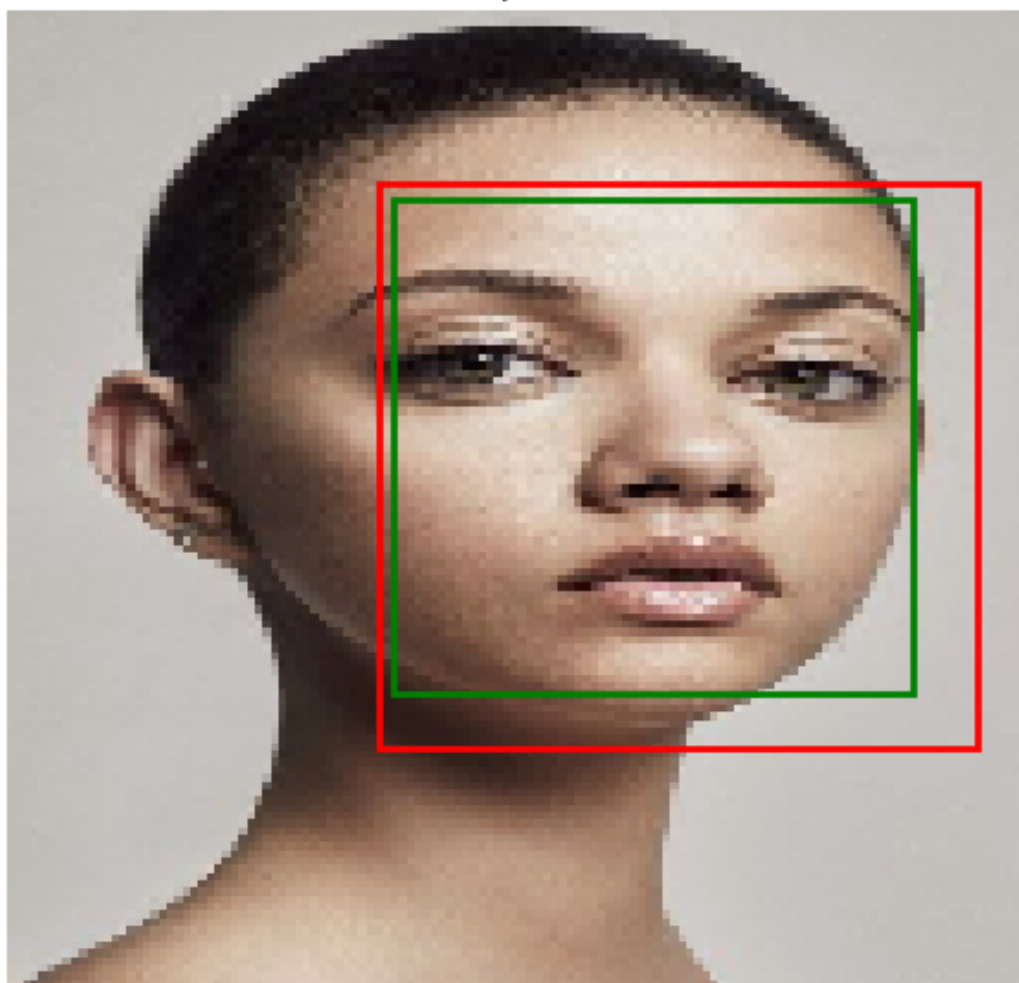
Plaintext: Model Prediction - Predicted Image

Green: Predicted



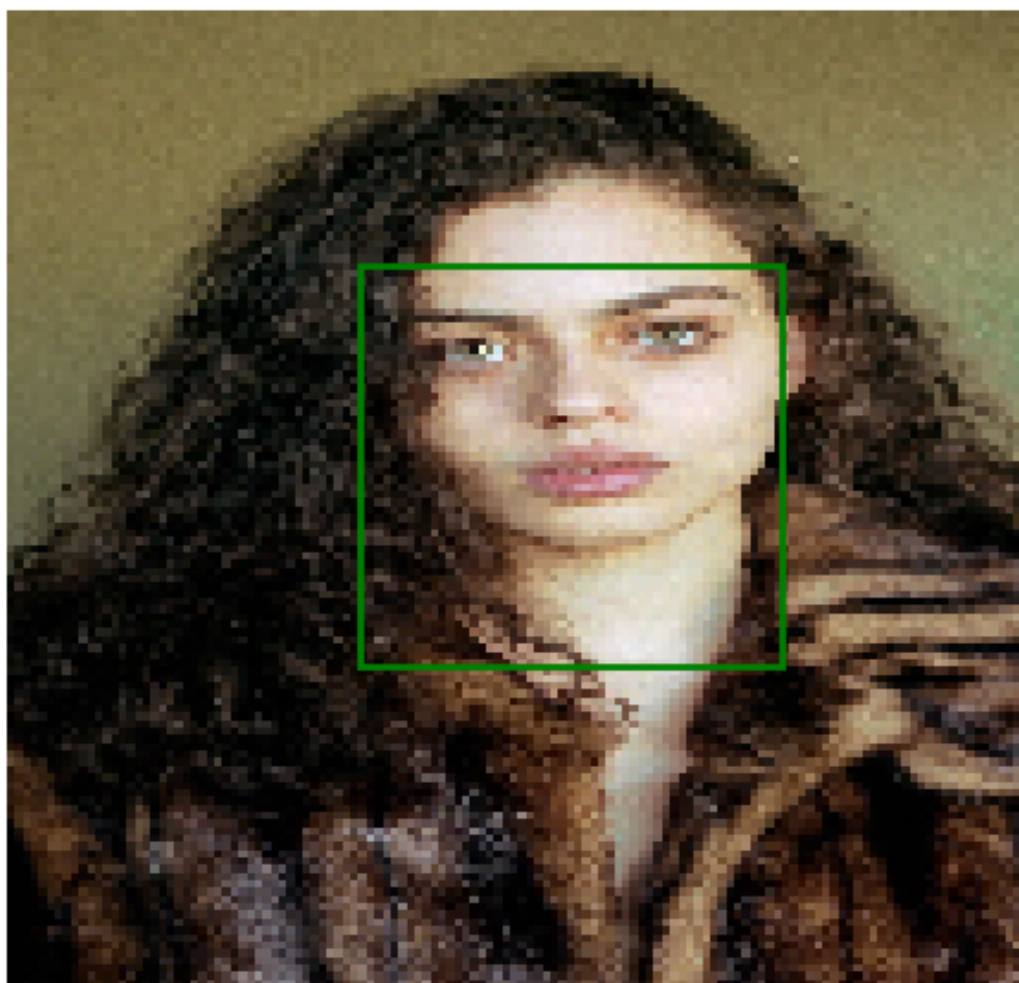
Plaintext: Model Prediction - Predicted Image

Green: Predicted | Red: Ground Truth



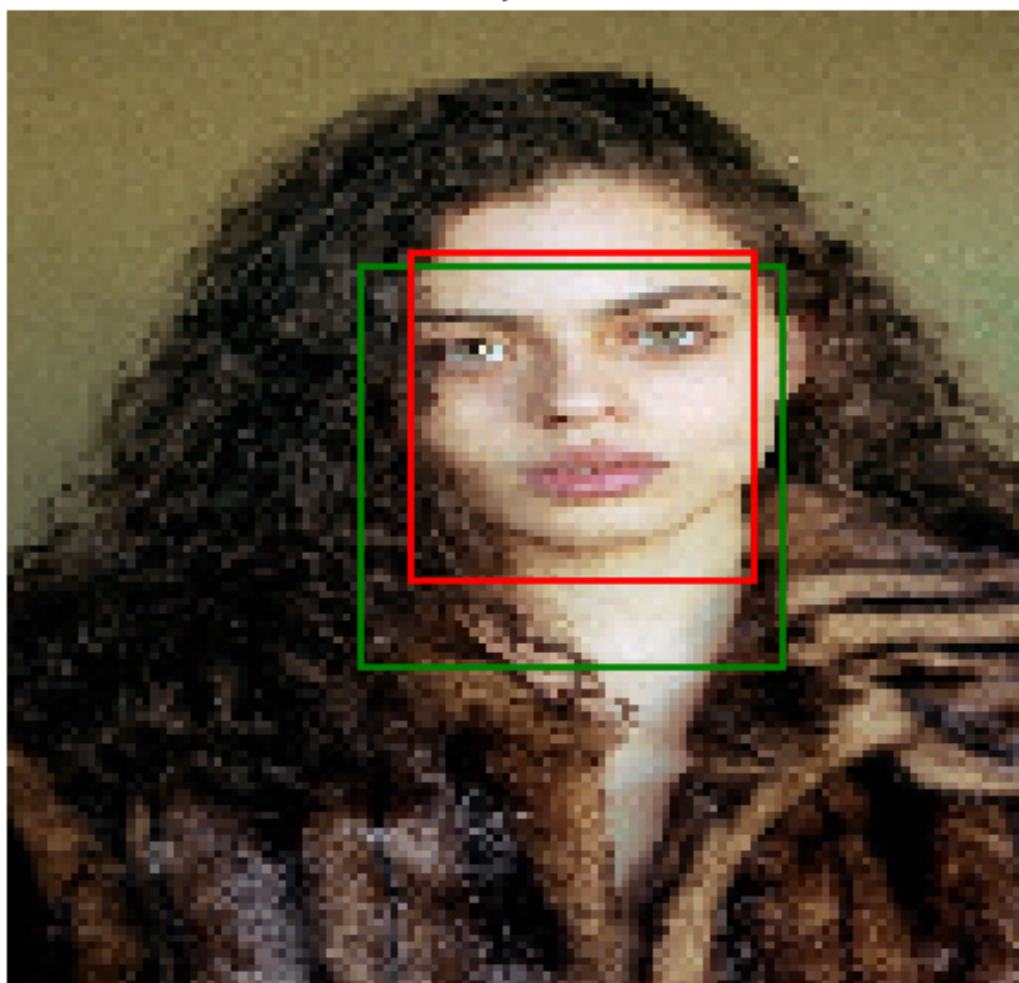
Plaintext: Model Prediction - Predicted Image

Green: Predicted



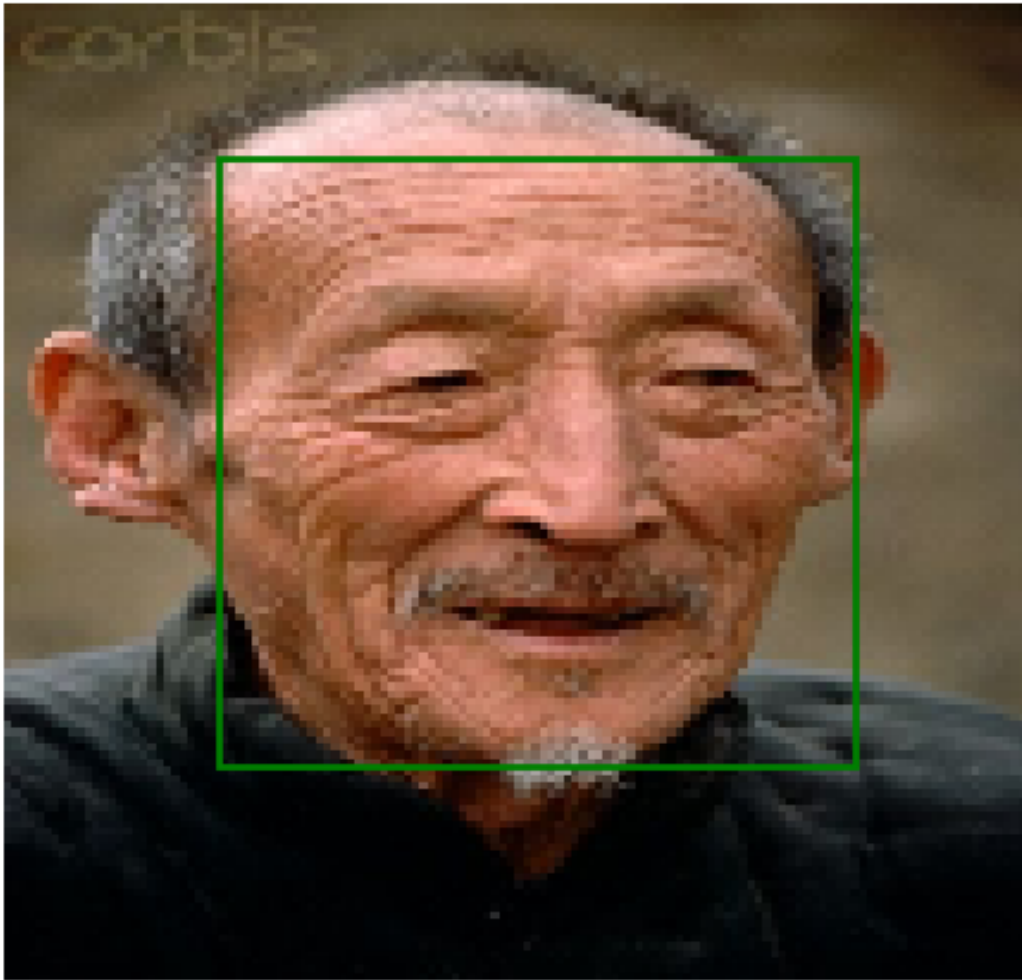
Plaintext: Model Prediction - Predicted Image

Green: Predicted | Red: Ground Truth



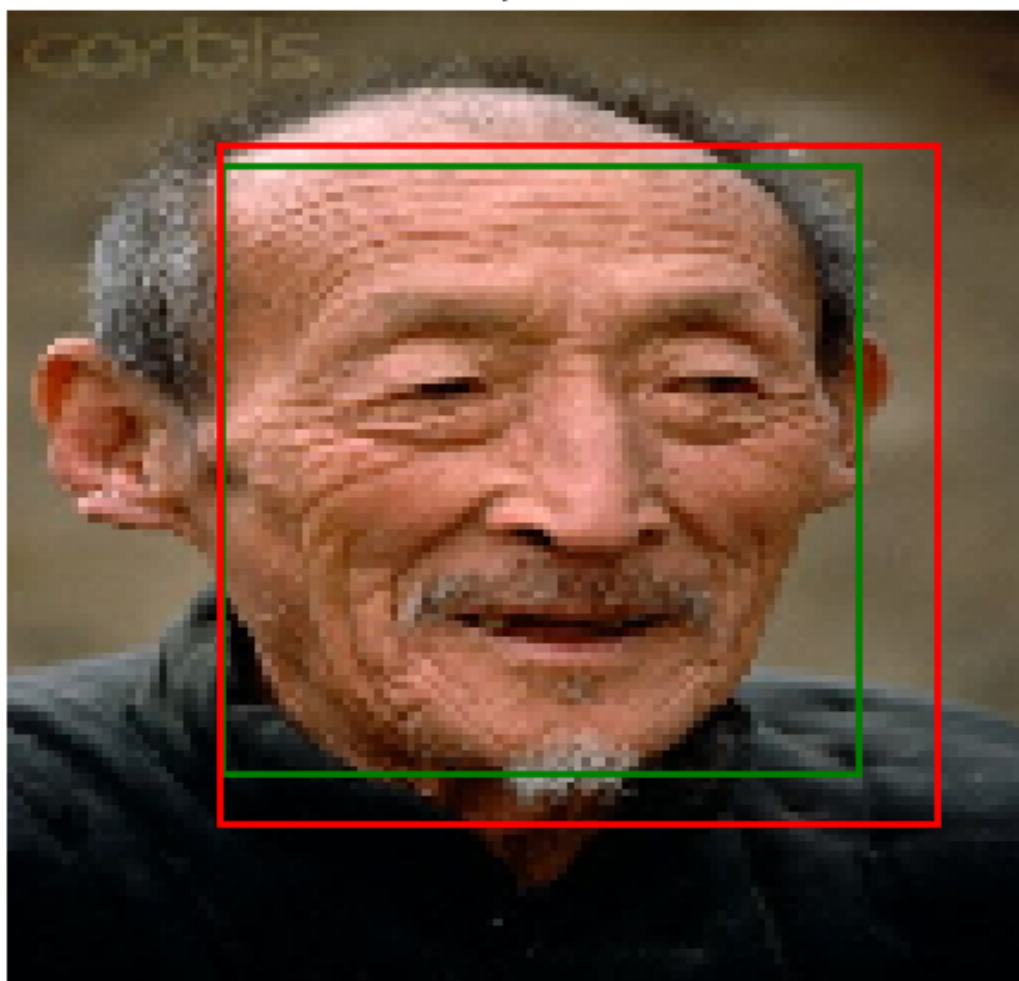
Plaintext: Model Prediction - Predicted Image

Green: Predicted



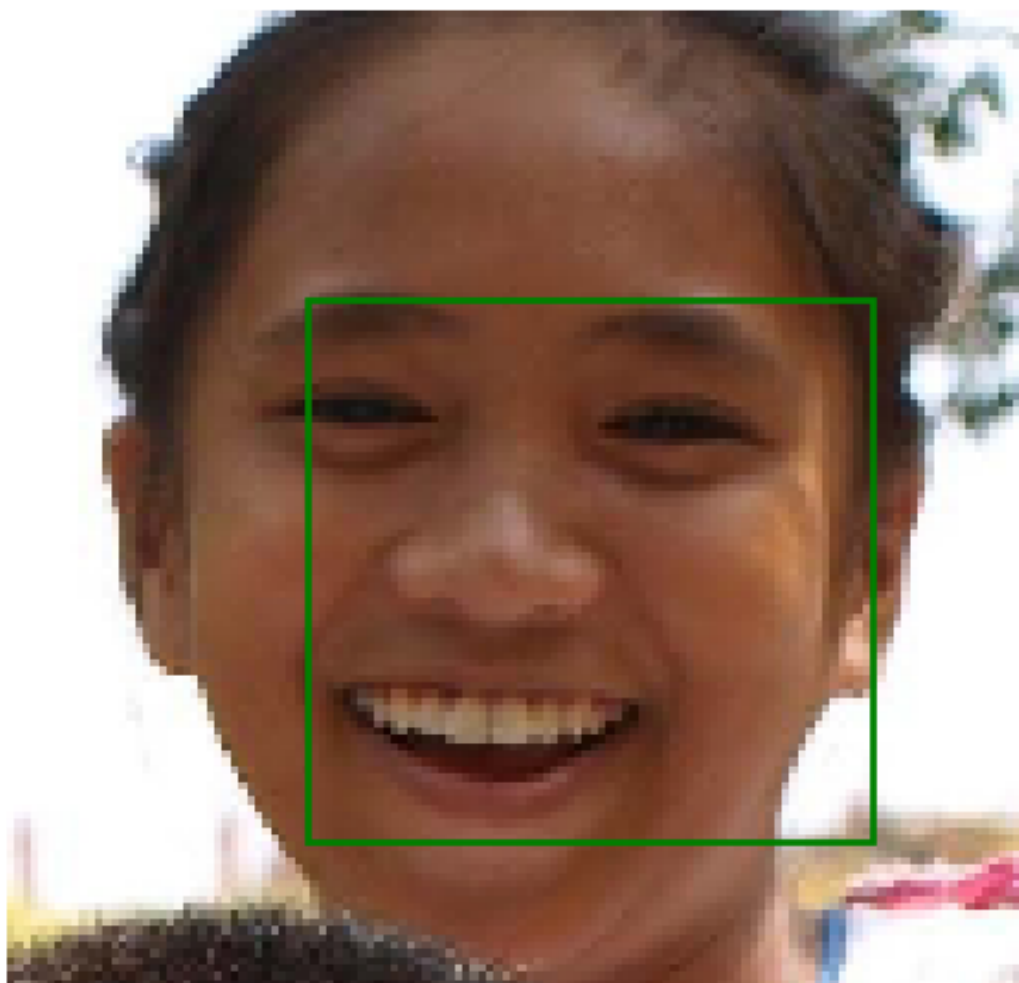
Plaintext: Model Prediction - Predicted Image

Green: Predicted | Red: Ground Truth



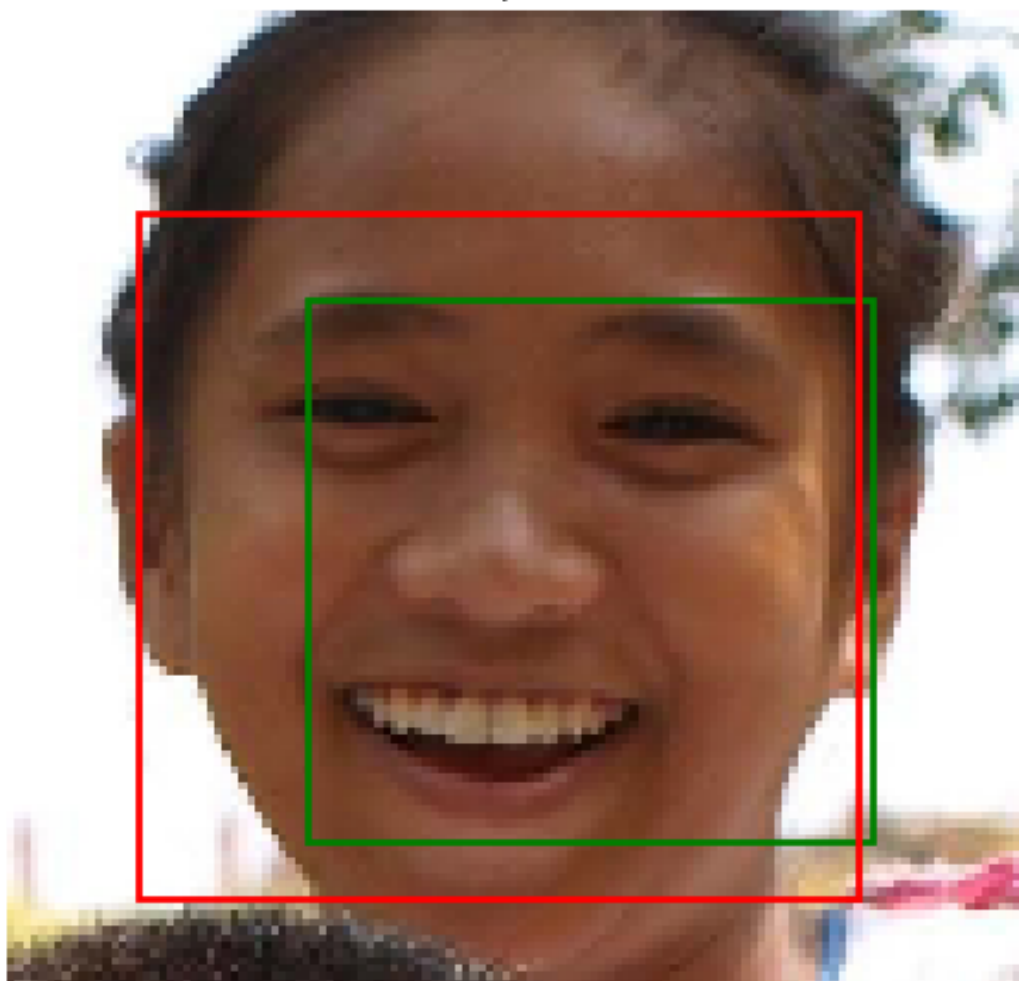
Plaintext: Model Prediction - Predicted Image

Green: Predicted



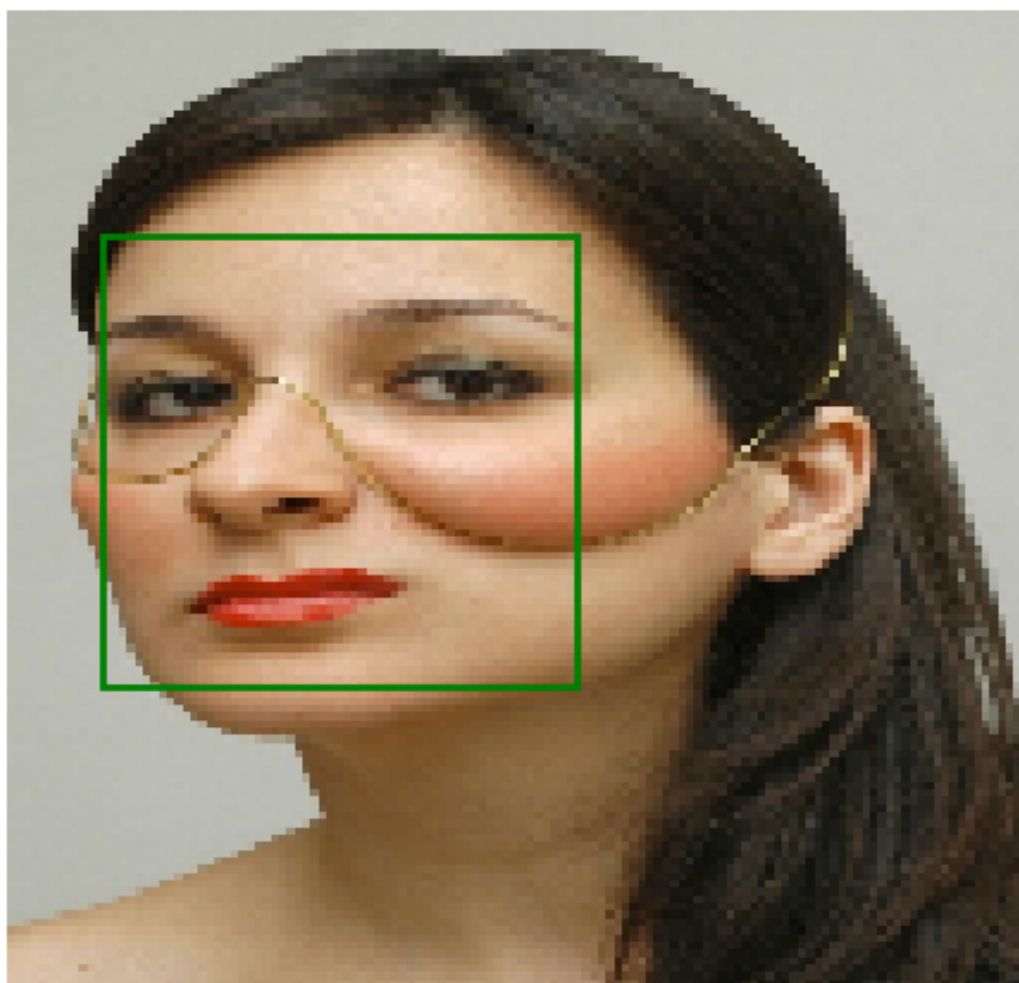
Plaintext: Model Prediction - Predicted Image

Green: Predicted | Red: Ground Truth



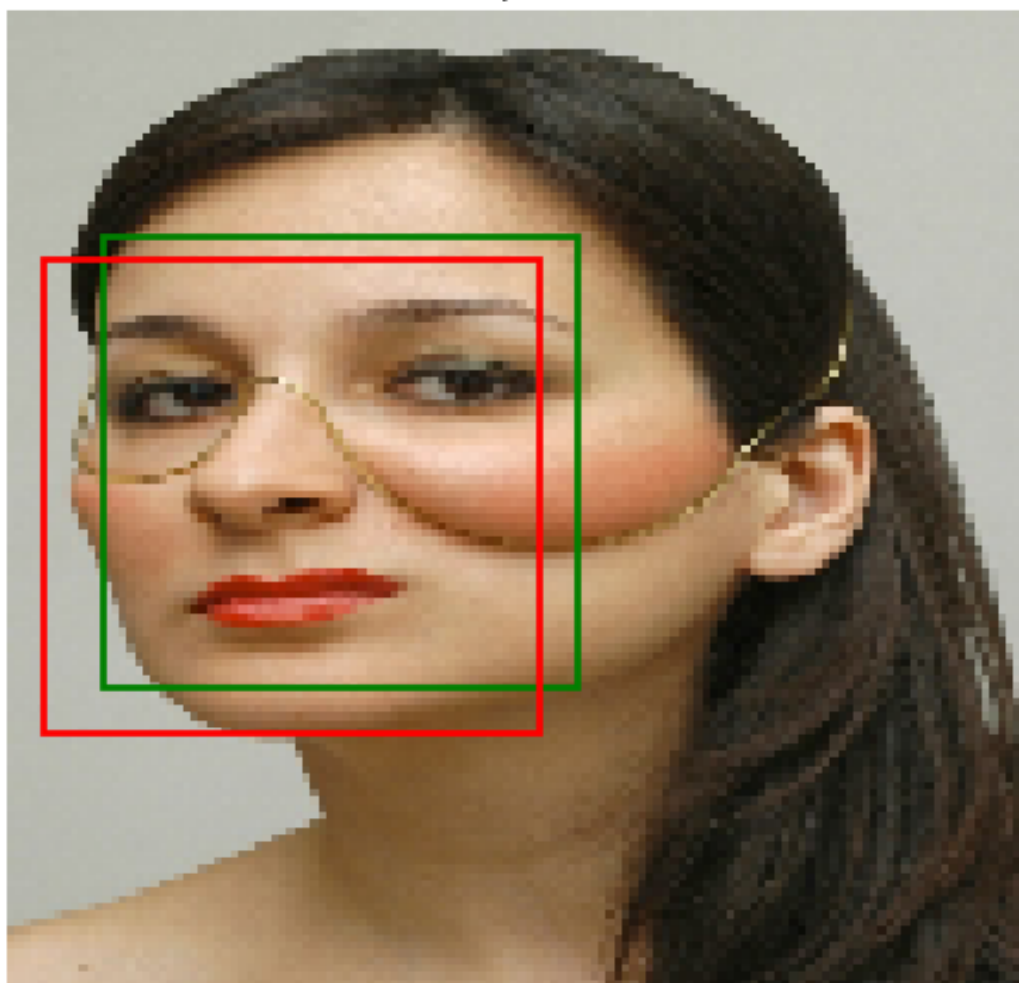
Plaintext: Model Prediction - Predicted Image

Green: Predicted



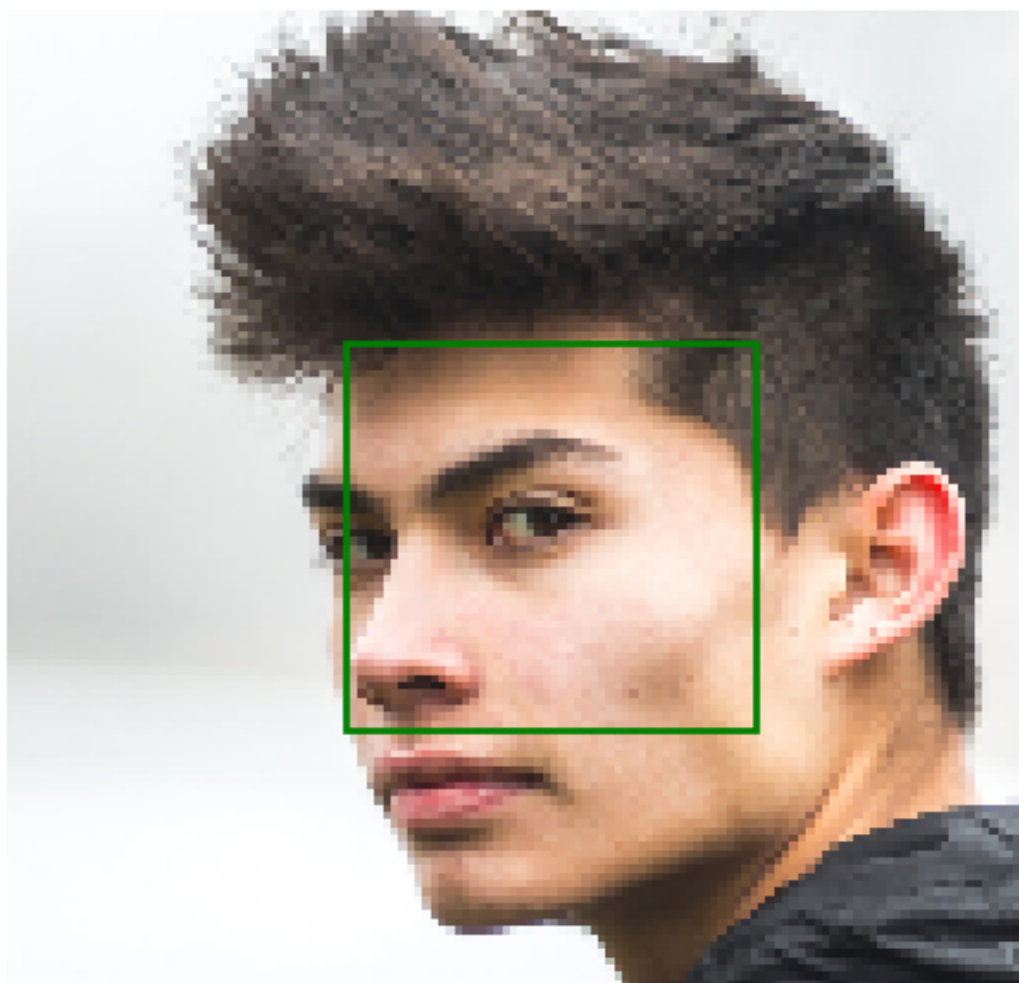
Plaintext: Model Prediction - Predicted Image

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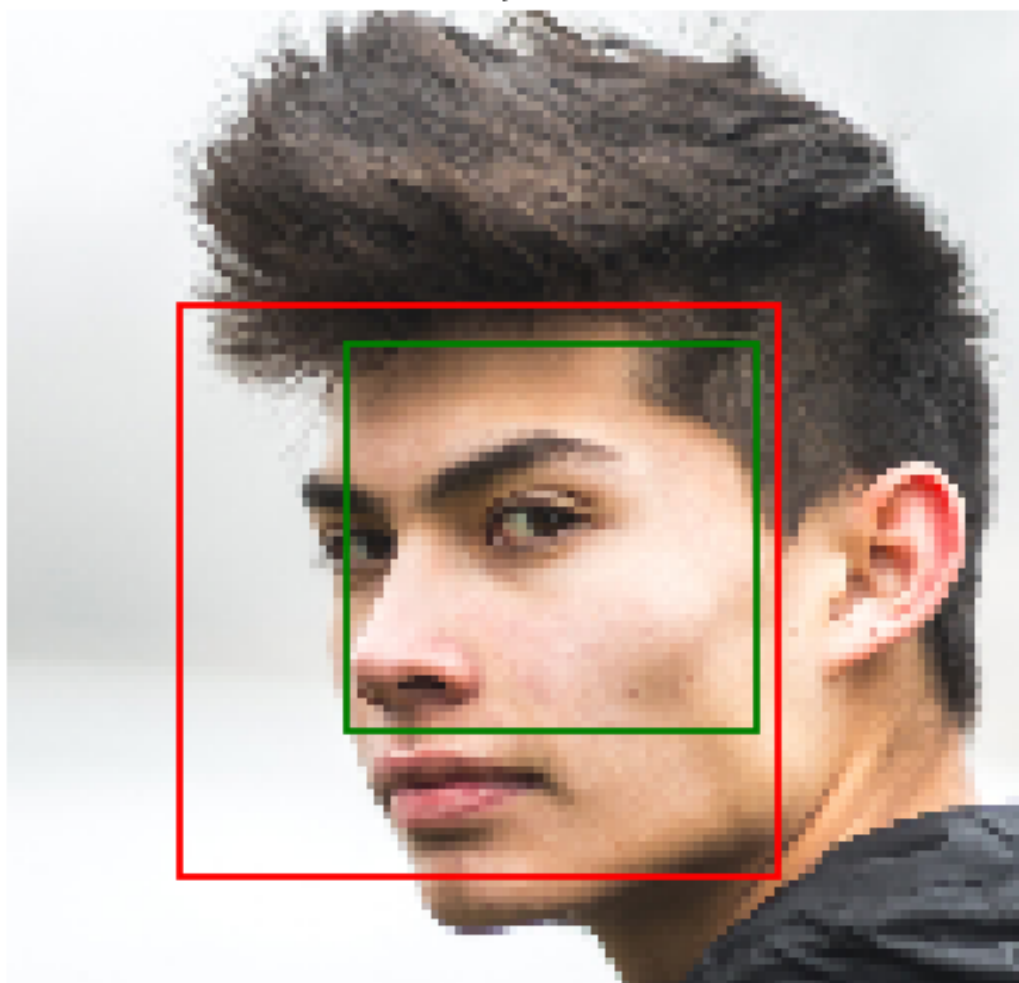
Plaintext: Model Prediction - Predicted Image

Green: Predicted



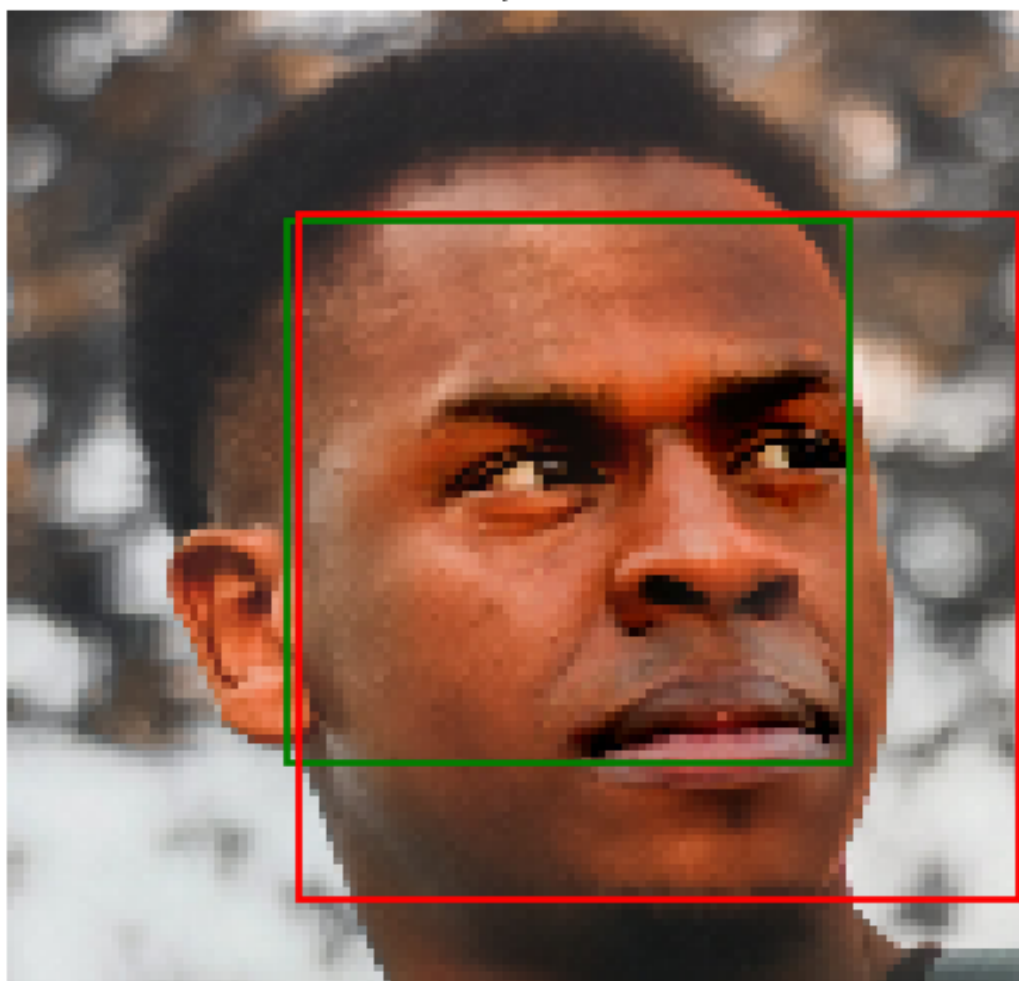
Plaintext: Model Prediction - Predicted Image

Green: Predicted | Red: Ground Truth



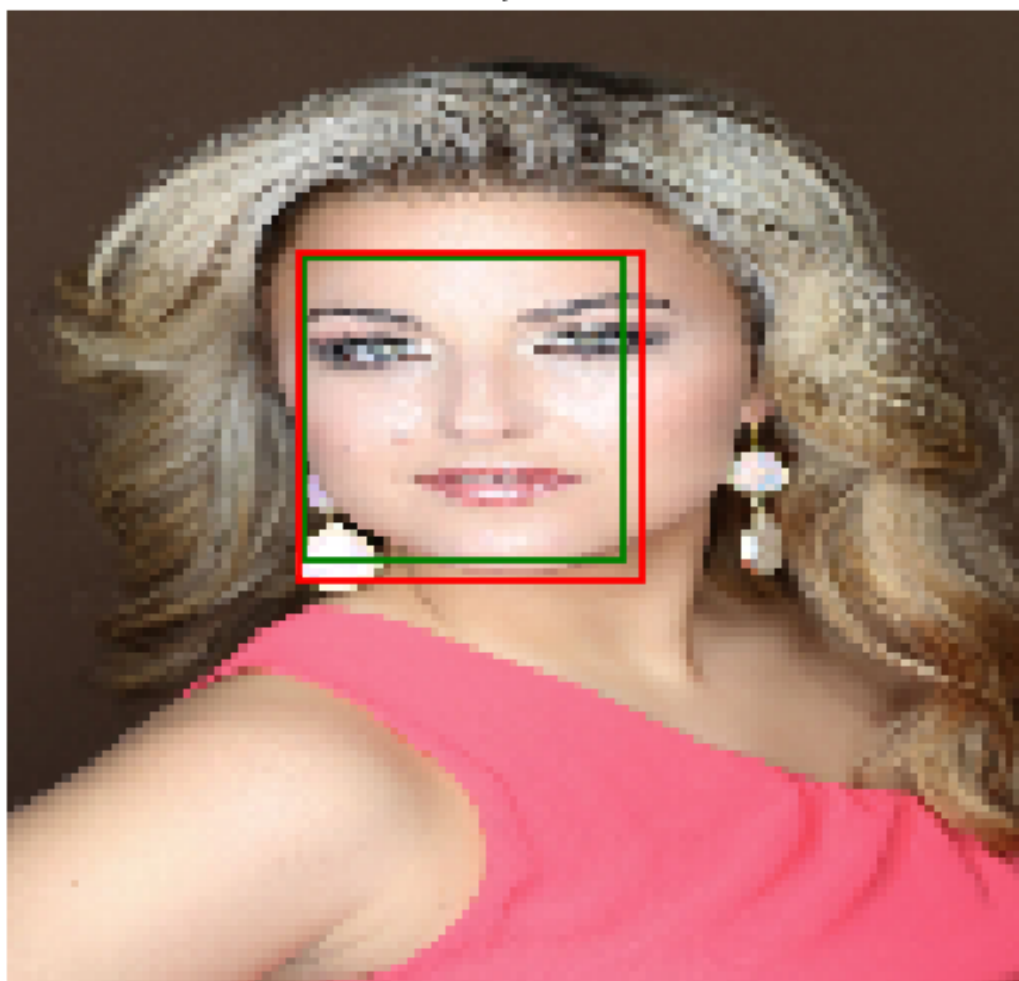
Encrypted: Model Prediction - Predicted Image

Green: Predicted | Red: Ground Truth



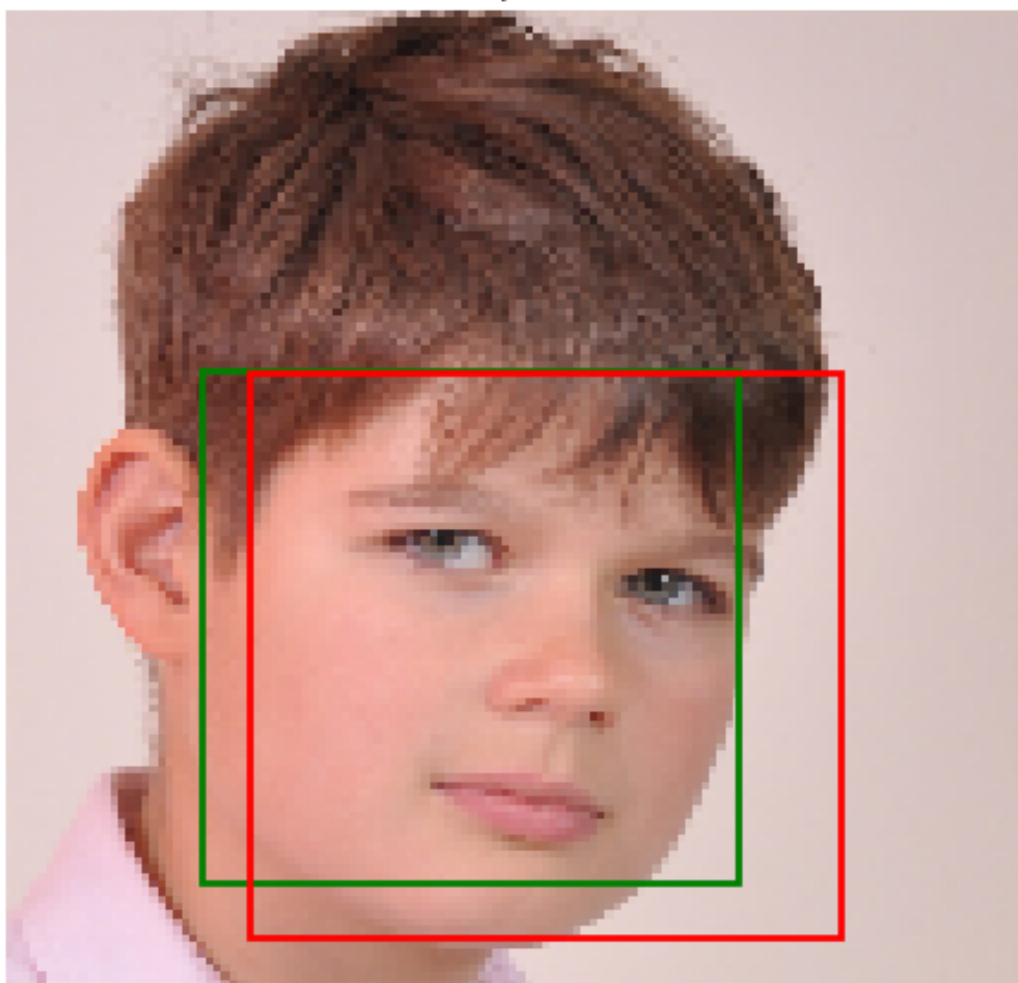
Encrypted: Model Prediction - Predicted Image

Green: Predicted | Red: Ground Truth



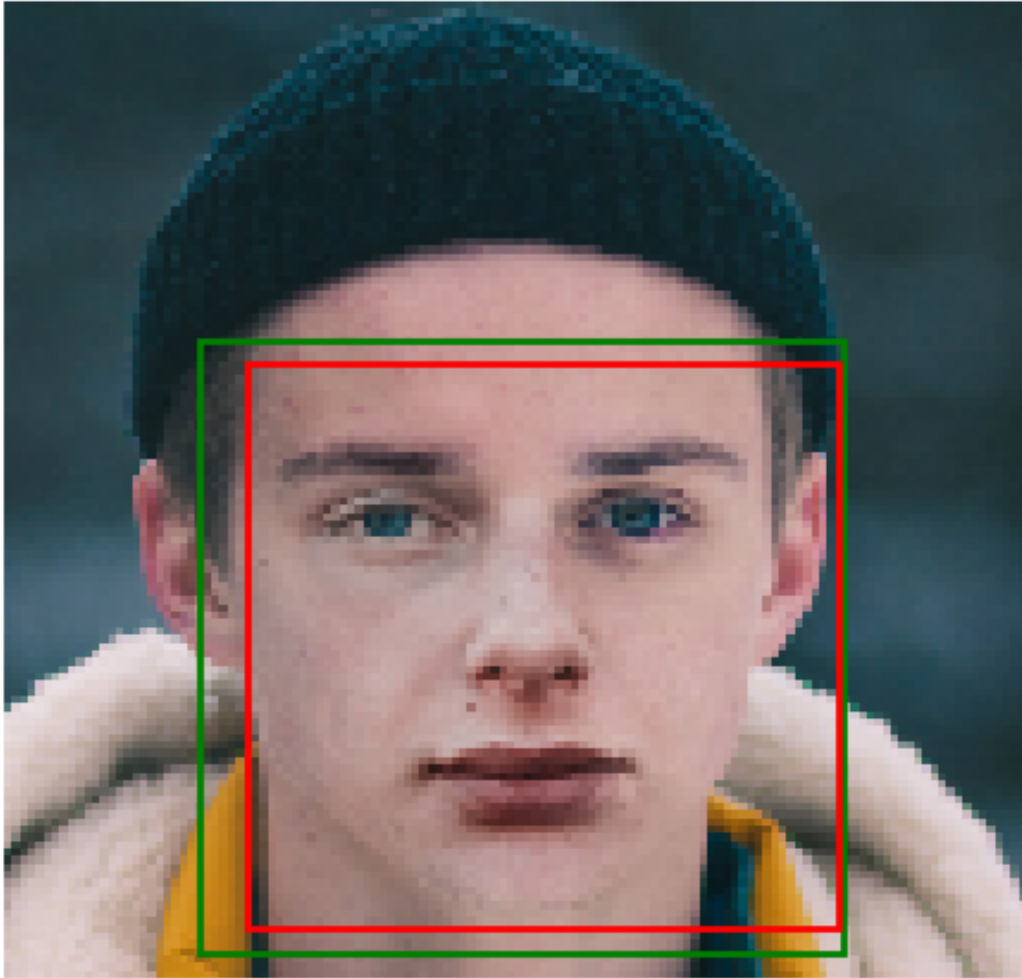
Encrypted: Model Prediction - Predicted Image

Green: Predicted | Red: Ground Truth



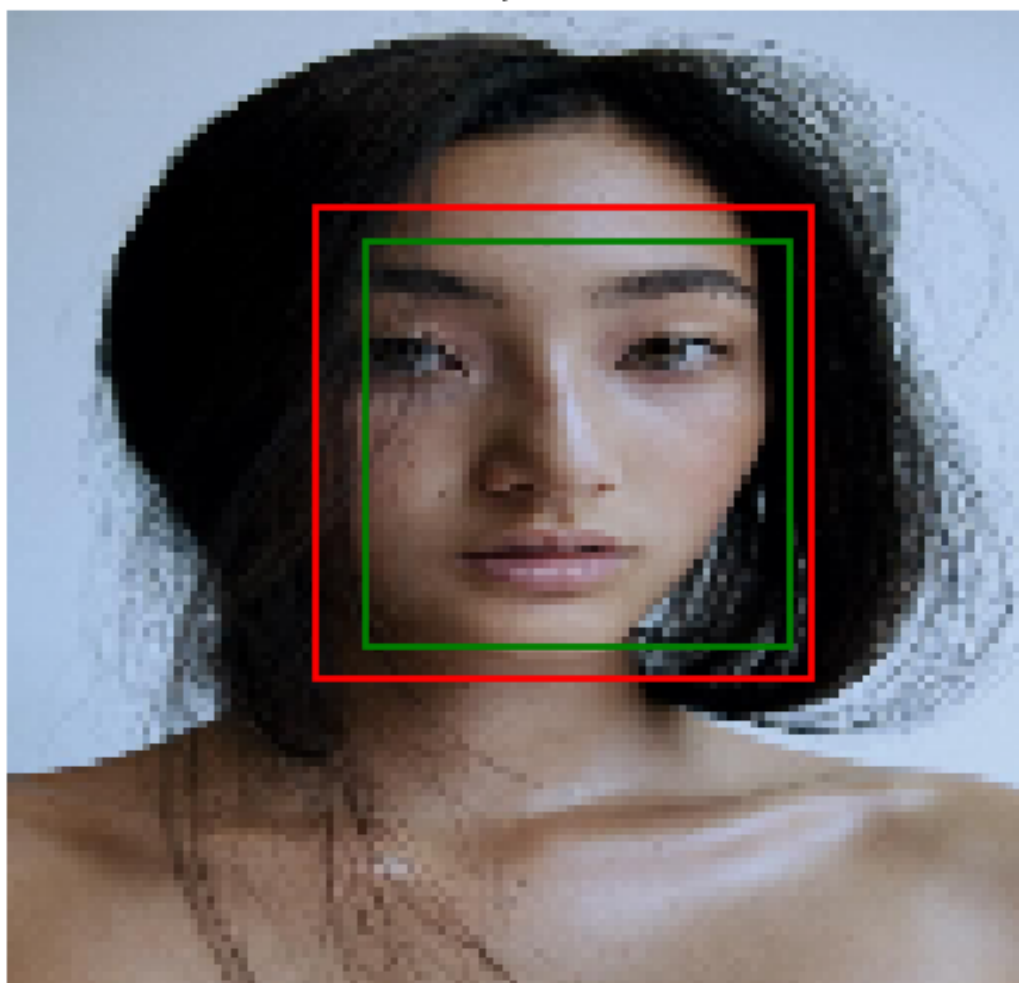
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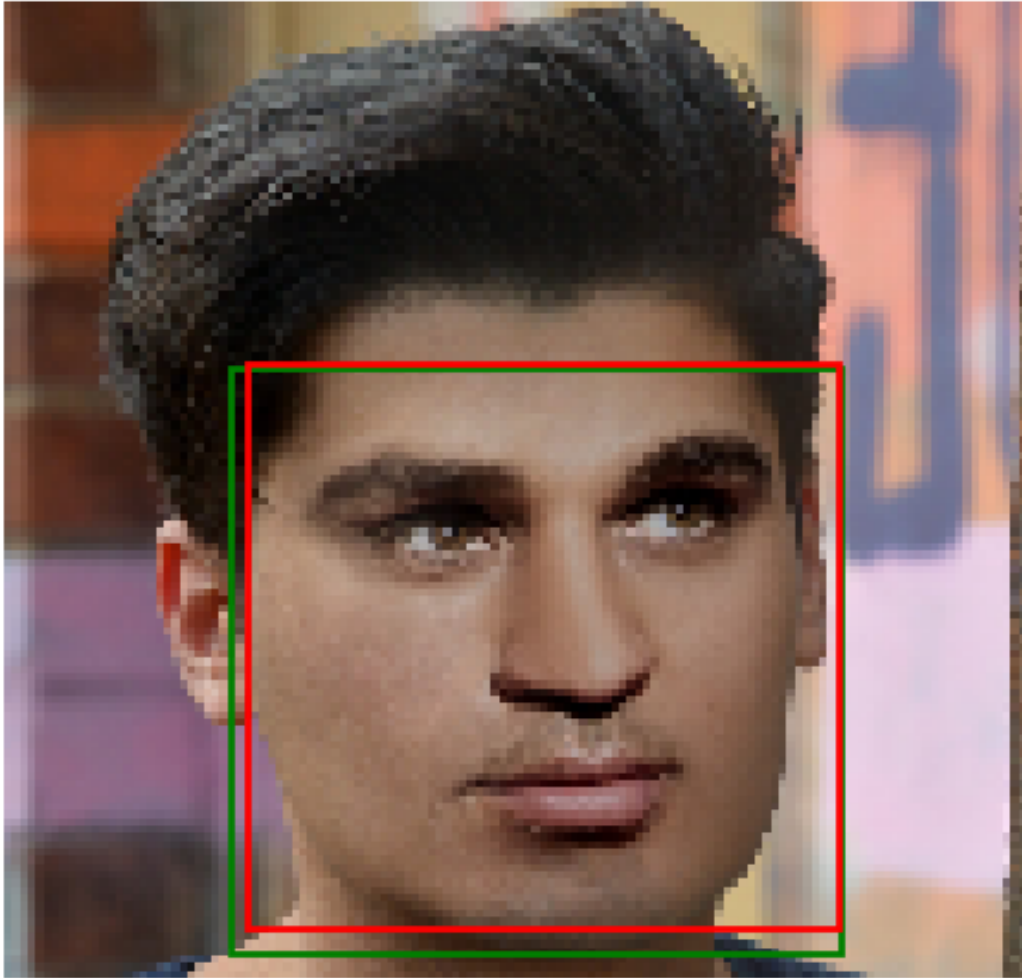
Encrypted: Model Prediction - Predicted Image

Green: Predicted | Red: Ground Truth



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Green: Predicted | Red: Ground Truth



Runtime System Resource Usage

encrypted-fhe: OS: macOS-26.0-arm64-arm-64bit
encrypted-fhe: Python Version: 3.10.8
encrypted-fhe: Processor: arm
encrypted-fhe: CPU Usage (%): 12.9
encrypted-fhe: RAM Usage (Used GB): 5.95
encrypted-fhe: RAM Usage (Total GB): 18.0
encrypted-fhe: RAM Usage (%): 77.9
encrypted-fhe: GPU: Not Available